## a Nanosheet gate-all-around flow (N3G, 18 EUV / 61 DUV masks)

| | | | | | |
|---|---|---|---|---|---|
| **1** STI + well trench isolation | **2** SiGe/Si superlattice epi | **3** fin cut + channel release | **4** inner spacer + S/D epi | **5** HKMG replacement gate | **6** contacts + MOL |
| **7** BEOL 12 metal layers | **8** BSPDN: wafer bond | **9** backside thinning + nTSV | **10** backside metal 3 layers | **11** hybrid bond L3 dies | **12** CoWoS + HBM attach |

Steps 1–6 are front-end (EUV critical layers at 2, 3, 5); 7–10 build the power delivery on the wafer backside; 11–12 are packaging.

## b Backside power delivery integration

| step | process | control |
|---|---|---|
| B1 | carrier wafer bonding | alignment < 0.5 $\mu$m |
| B2 | Si thinning to 400 nm | TTV < 1 $\mu$m |
| B3 | nTSV 0.4 $\mu$m pitch | Rc < 1 Ω |
| B4 | 3 backside metals | IR drop < 50 mV |
| B5 | backside passivation | leakage < 1 nA |

### Why backside power

- front-side tracks freed for signal: +5.4% routed area
- IR drop 62 → 48 mV at 80 A
- via resistance 1.8 → 0.9 Ω
- 6% lower dynamic power at iso-clock

BSPDN is what makes a 0.75 V, 80 A, 65 W part practical in a two-slot card: the front-side grid alone would have needed 14 metal layers.